

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – MOCK INTERVIEW (20 Questions)

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Naturalization(BL5)	Assessment:	Assessment Tool 2 – Mock Interview
Weightage:	20% of CIA	Total Marks:	20 (20 Questions × 1 Mark)
CO4 Statement:	<i>Evaluate visualization designs based on perception, cognition, and effectiveness principles (BL5).</i>		
Student Name:	Monesh S	Reg. No.:	192324027
Section / Batch:	B.Tech (AI&DS)	Date:	09/08/2026

Code of Conduct

I Monesh S / 192324027 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Monesh S

Instructions

- This test contains 20 questions, each carrying 1 mark. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus Area	No. of Questions	Marks
Scatterplots & Bubble Charts	Scatterplots and bubble charts	2	2
Correlation Analysis	Correlogram and feature selection	2	2
Dimensionality Reduction Techniques	PCA, t-SNE, UMAP	3	3
Density & Overplotting Techniques	Hexbin plots and overplotting handling	2	2
Network Visualization	Nodes and edges in network graphs	2	2
Time Series Analysis	Moving average, detrending, smoothing, seasonality, noise	5	5
Geospatial Visualization	Geospatial plots, map projections, choropleth maps, spatial heatmaps	4	4
GRAND TOTAL:		20 Marks	

1. What is the purpose of a histogram?

A histogram shows the frequency distribution of numerical data using bars.

2. How does a box plot help identify outliers?

Values outside the lower and upper whiskers are usually considered outliers.

3. What information does a violin plot provide?

It shows the distribution, density, spread, median, and shape of the data.

4. What is the difference between a bar chart and a histogram?

A bar chart represents categorical data, while a histogram represents the distribution of numerical data.

5. Why are heatmaps useful in data analysis?

Heatmaps make patterns, relationships, and high or low values easy to identify using colors.

6. What does color intensity represent in a heatmap?

Color intensity represents the magnitude or value of the data.

7. What is a pair plot used for?

A pair plot is used to study relationships and correlations between multiple variables.

8. How can a density plot help understand data distribution?

It shows the shape and concentration of continuous data using a smooth curve.

9. What is meant by dimensionality reduction?

Dimensionality reduction means reducing the number of variables or features while retaining important information.

10. Why is PCA commonly used before visualization?

PCA reduces many dimensions into a smaller number of components, making high-dimensional data easier to visualize.

11. What is a principal component?

A principal component is a new variable that captures the maximum possible variation in the data.

12. How does UMAP preserve data structure?

UMAP preserves important local relationships and some global structure when reducing high-dimensional data.

13. What is the main goal of t-SNE visualization?

The main goal of t-SNE is to visualize similar data points close together and different points farther apart.

14. What does a contour plot represent?

A contour plot represents three-dimensional data on a two-dimensional surface using contour lines.

15. How are contour lines interpreted in a contour plot?

Each contour line connects points having the same value.

16. What is a radar chart used for?

A radar chart is used to compare multiple variables or characteristics of different objects.

17. What does each axis in a radar chart represent?

Each axis represents one variable or performance measure.

18. How does a parallel coordinates plot visualize multivariate data?

It uses multiple vertical axes, with each line representing one observation across different variables.

19. What is overplotting in data visualization?

Overplotting occurs when too many data points overlap, making the visualization difficult to understand.

20. Name one visualization technique used for high-dimensional data.

PCA is one technique commonly used to visualize high-dimensional data.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Concepts	50%	Demonstrates thorough and accurate understanding of concepts	Shows good understanding with minor errors	Basic understanding with limited clarity	Poor understanding or incorrect responses		
Critical Thinking & Response	50%	Provides logical, confident, and well-structured responses	Gives relevant responses with minor gaps	Responses are partially correct with limited analysis	Unable to analyse or respond effectively		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____